ICGINet: Intratemporal and Cross-Temporal Graph Interaction Network for Misaligned Bitemporal Remote Sensing Image Change Detection

Kexin Zhu, Jun Pan, Yang Yang, Yuchuan Bai, *Student Member, IEEE*, Rui Xu, *Graduate Student Member, IEEE*, and Junli Li

***Abstract*—Change detection (CD) identifies pixel-level land cover changes in multitemporal remote sensing imagery (RSI), playing a critical role in environmental monitoring and disaster management. Despite the advantages of high-resolution imagery in providing detailed spatial and contextual information, it significantly intensifies the complexity of fine-grained CD by exacerbating sensitivity to spatial misalignment. While accurate CD conventionally hinges on precise image registration, inevitable misalignments in practical scenarios often lead to spurious change responses and compromise the stability of feature representation learning. To address these challenges, this article proposes the intratemporal and cross-temporal graph interaction network (ICGINet), a novel architecture that enhances CD robustness under registration uncertainty by modeling intratemporal and cross-temporal dependencies through adaptive graph-based message passing and relational reasoning. Specifically, ICGINet employs a dual-branch design that combines convolutional neural networks (CNNs) for local semantic extraction with graph neural networks (GNNs) for global relational modeling, enabling the integration of fine-grained spatial details and long-range contextual dependencies. To reconcile intratemporal features from these heterogeneous branches, a dual-focus feature fusion (DFFF) module is proposed to facilitate attention-guided information exchange and semantic alignment. Furthermore, an adaptive neighborhood graph awareness (ANGA) module constructs a displacement-aware dynamic graph to capture cross-temporal neighborhood correspondences, thereby mitigating spatial misalignment during temporal interaction. Comprehensive experiments on two challenging datasets demonstrate that ICGINet achieves superior performance compared to state-of-the-art methods, both in terms of accuracy and robustness to registration error.**

***Index Terms*—Change detection (CD), graph convolutional network (GCN), image registration, remote sensing, spatial alignment.**

Received 22 May 2025; revised 18 August 2025; accepted 8 September 2025. Date of publication 11 September 2025; date of current version 29 September 2025. This work was supported by the National Natural Science Foundation of China under Grant 62371352 and Grant 62425102. *(Corresponding author: Jun Pan.)*

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Digital Object Identifier 10.1109/TGRS.2025.3608469

I. Introduction

CHANGE detection (CD) is a technique used to identify land cover or land use changes over time by comparing multitemporal remote sensing imagery (RSI) of the same area [1]. By filtering out irrelevant noise and highlighting meaningful changes within a specific period, CD has been widely applied in various fields, including urban planning, disaster monitoring, land cover change analysis, and map updating [2], [3], [4]. As spatial resolution in RSI improves, high-resolution imagery captures more details, enabling more accurate and insightful CD analysis, which has therefore emerged as a key focus in remote sensing research [5], [6], [7]. Nevertheless, the improvement in spatial resolution also renders CD more susceptible to image misregistration, thereby potentially compromising its overall performance and reliability in practical applications [8], [9].

Conventional CD approaches can be broadly classified into three categories: pixel-based, feature-based, and classification-based methods. Pixel-based approaches primarily operate at the per-pixel level using arithmetic operations like image differencing and ratioing, offering computational efficiency but being prone to noise [10]. Feature-based methods extract more representative and discriminative features (e.g., texture, shape) to reduce noise, although feature selection remains a challenging and experience-driven process [11], [12]. Classification-based methods independently classify multitemporal images and compare the resulting labels, generally achieve higher detection accuracy but being sensitive to classification accuracy, leading to error accumulation and reduced reliability [13], [14]. Overall, conventional methods encounter several challenges in CD, including difficulties in robust feature extraction, sensitivity to noise, and limited capacity to model nonlinear change patterns [15].

With the rise of deep learning in remote sensing applications, convolutional neural networks (CNNs) have become a predominant architecture for CD, owing to their strong capacity for hierarchical feature extraction [16], [17], [18]. However, CNNs inherently suffer from limited receptive fields, which constrain their ability to capture long-range dependencies. To address this issue, researchers have explored strategies such as increasing network depth [19] and employing dilated

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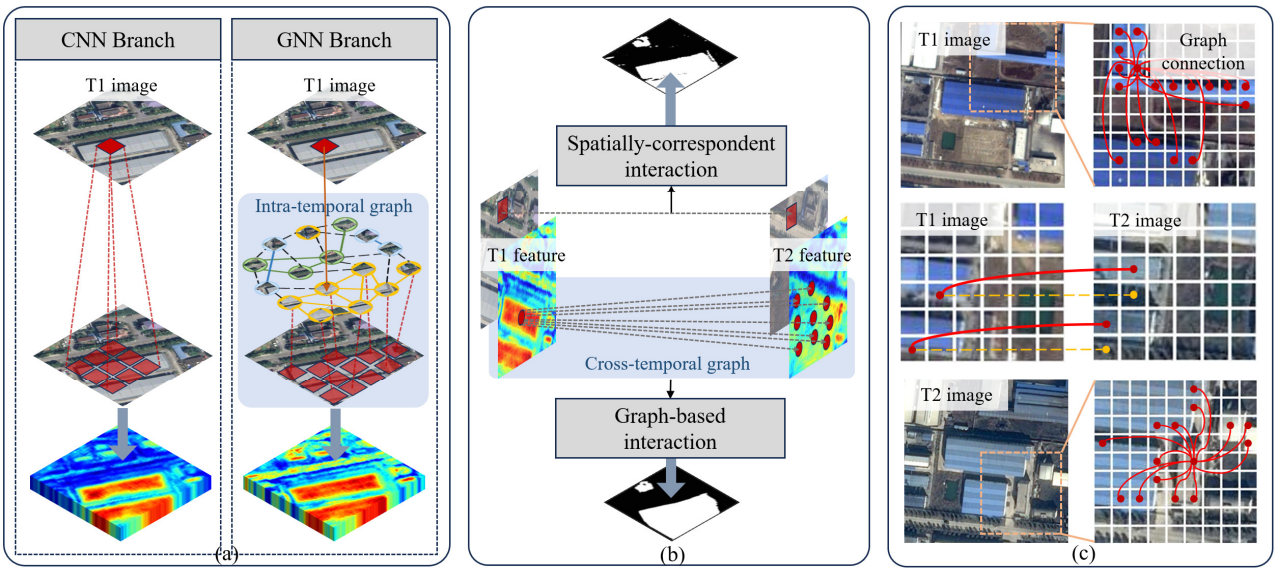


Fig. 1. Comparative visualization of feature extraction and interaction mechanisms for the proposed intratemporal and cross-temporal graph structure. (a) Comparison between conventional CNN-based feature extraction and the GNN-based approach adopted in this article for intratemporal feature extraction. (b) Comparison between conventional spatially-correspondent interactions and the proposed graph-based interaction method for cross-temporal feature interaction. (c) Visualization of intratemporal and cross-temporal graph interactions in a specific example.

convolutions to enlarge the receptive field [20], aiming to improve the ability of CNNs to capture global contextual dependencies. In parallel, attention mechanisms have been introduced to indirectly enhance the receptive field quality by enabling the network to selectively focus on change-relevant features across spatial locations [21], [22], [23]. Nevertheless, these approaches still have room for improvement in capturing long-range dependencies. The recent success of transformers in computer vision (CV) has opened new avenues for CD [24], [25], [26]. By leveraging the self-attention mechanism, transformers substantially improve their capacity to model long-range dependencies and ensure more coherent and accurate feature representations [7], [27], [28], [29], [30]. However, applying transformers to VHR RSI-CD remains challenging due to their high computational cost and inefficiency.

Beyond advancements in network architecture design, an equally critical challenge in CD lies in its strong dependence on precise bitemporal image registration. Although CNN-based and transformer-based methods have significantly improved CD accuracy over traditional techniques—effectively addressing issues such as feature ambiguity and class imbalance—they still fundamentally rely on accurately aligned image pairs, typically requiring misalignment to be within 0.5 pixels [31]. However, in real-world scenarios, acquiring bitemporal images that satisfy this stringent criterion is particularly challenging, which in turn undermines the reliability and accuracy of CD [32]. This issue becomes even more pronounced in practical deployment settings, such as onboard remote sensing satellites, where computational resources are inherently constrained. In such contexts, accurate image registration is not only difficult to achieve but also computationally demanding and time-intensive, thereby imposing substantial limitations on real-time image analysis and decision-making capabilities. In response to the challenges posed by misregistration, recent studies have proposed joint optimization frameworks that unify image registration and CD into a unified pipeline, enabling integrated execution while achieving satisfactory accuracy and efficiency [33], [34]. While promising, these methods still substantially rely on precise registration, which may affect their robustness and practicality. Beyond direct registration integration, alternative alignment strategies have been explored, including geometric deformation field estimation for feature-level alignment [35], [36], or improving model robustness to misregistration via techniques such as deformable convolutions [37], [38], thereby alleviating the need for explicit registration.

Graph structures provide a flexible representation of complex, non-Euclidean data. Compared to CNNs and transformers, which use regular grids or sequences to represent images, graph neural networks (GNNs) are better suited for capturing irregular shapes and intricate semantic information within images [39]. Although GNNs were initially not designed for 2-D image processing, the spatially adapted graph convolutional networks (GCNs) proposed by Micheli [40] and its subsequent variants have successfully extended the application of graph structures in CV [41], [42], [43], achieving remarkable progress in recent remote sensing studies [44], [45], [46]. Through the learning of node embeddings and the use of a message passing mechanism for feature aggregation, GCNs are capable of capturing long-range dependencies and offer a more flexible receptive field compared to traditional CNNs [Fig. 1(a)]. Recent studies have applied GCNs to CD tasks across diverse RS modalities—such as high-resolution [47], [48], [49], [50], hyperspectral [51], [52], and SAR imagery [53]—demonstrating their ability to model hierarchical spatial dependencies and learn discriminative representations. This message-passing mechanism in graph-based models also provides a novel solution for CD methods that are robust to registration error. Unlike traditional spatially-correspondent interaction schemes, which heavily rely on precise spatial alignment—such as pixel-wise concatenation or differencing of bitemporal features—the graph-based interaction scheme facilitates feature propagation by leveraging edge relations to

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establish connections between semantically similar nodes [Fig. 1(b)] [54]. This property alleviates the adverse effects of pixel shifts, which typically compromise feature learning stability in registration-dependent methods [35]. Moreover, the graph-based interaction mechanism exploits edge features to model cross-pixel dependencies, circumventing the need for precise spatial alignment and potentially enhancing robustness to pixel misalignment. As illustrated in Fig. 1(c), the intratemporal graph links nodes through semantic representations, aiding in more accurate object identification, while the cross-temporal graph overcomes spatial constraints (yellow lines indicate spatial correspondences) to establish connections between semantically similar nodes (red lines indicate semantic correspondences).

To address the critical challenge of CD in misaligned RSI, this article proposes ICGINet: an intratemporal and cross-temporal graph interaction network (ICGINet) tailored for robust CD under spatial misalignment. By leveraging the flexibility and relational reasoning capabilities of graph structures, ICGINet adaptively constructs intratemporal and cross-temporal interactions that not only strengthen long-range dependency modeling but also enhance representation learning, while mitigating the adverse effects of spatial misalignment on feature learning and detection accuracy. Specifically, the proposed method introduces a dual-branch architecture comprising a CNN-based semantic extractor and a GNN-based relational encoder: the former captures rich local details and texture patterns, while the latter models global contextual dependencies and semantic relationships. Nevertheless, the inherent representational gap between the CNN and GNN branches gives rise to semantic inconsistency, which is mitigated through the incorporation of a dual-focus feature fusion (DFFF) module. This module serves as an inter-branch communication conduit, facilitating mutual calibration between localized detail extraction and global structural encoding within each temporal phase. Building upon this harmonized intratemporal representation, the adaptive neighborhood graph awareness (ANGA) module further refines cross-temporal dependencies. In contrast to conventional pixel-wise interaction strategies that rely on rigid spatial correspondences, ANGA constructs a dynamically connected graph structure that adaptively models spatial displacements between bitemporal features, thereby enhancing the model's ability to capture complex change patterns under misaligned conditions. The main contributions of our work are as follows.

1) A novel graph-based network, ICGINet, is proposed to enhance the robustness of CD under spatial misalignment by modeling intratemporal and cross-temporal dependencies through adaptive graph interactions.

2) The DFFF module is designed to perform attention-driven graph-based feature fusion to reconcile semantic inconsistencies between dual branches within each temporal phase.

3) The ANGA module is proposed to build a cross-temporal graph that dynamically captures spatially displaced neighborhood relationships, thereby mitigating the effects of registration error on temporal consistency.

II. Methodology

A. Overall Structure of ICGINet

Motivated by the limitations of conventional pixel-wise classification methods in handling spatial misalignment and the growing success of graph-based models in capturing complex relationships, this article proposes ICGINet, a graph-based interaction architecture tailored for robust CD. Unlike traditional CNN-based architectures that rely on fixed grid structures and struggle with long-range dependencies and spatial inconsistencies, graph-based models offer a flexible and adaptive framework capable of modeling non-Euclidean structures and semantic relations across spatial and temporal domains. This makes them particularly well-suited for CD tasks under misaligned conditions, where explicit pixel-level correspondence is unreliable or unavailable. Guided by these insights, ICGINet is designed with three key components: a dual-branch structure consisting of a semantic extractor and a relational encoder, a DFFF module for intratemporal semantic alignment and fusion, and an ANGA module for cross-temporal graph construction and interaction.

Fig. 2 provides a visual overview of the general architecture of ICGINet. Given a pair of pre-registered bitemporal images I_{t1} and I_{t2}, a random perspective transformation is applied to the pre-change image I_{t1} to simulate realistic registration error, resulting in a warped image I'_{t1}. This image, along with the post-change image I_{t2}, is then fed into a dual-branch network consisting of a CNN-based semantic extractor and a GNN-based relational encoder. Both branches share weights across the two temporal inputs to ensure temporal consistency. After passing through the dual branches, each temporal input yields a pair of feature maps: F_{ct1} and F_{gt1} for I'_{t1}, and F_{ct2} and F_{gt2} for I_{t2}. However, due to the heterogeneous nature of CNN and GNN representations, semantic inconsistency may arise, potentially hindering effective fusion. To reconcile this, the DFFF module is introduced to integrate intratemporal features at each time phase. In DFFF, CNN features are first projected into a graph-compatible space and then fused with GNN features through attention-guided message passing, resulting in harmonized representations F_{t1} and F_{t2} that encode both local details and global structure. The fused features are then passed into the ANGA module, which constructs a cross-temporal graph to explicitly model displaced spatial correspondences. These dynamic edges encode spatial displacements and facilitate edge-aware message passing, allowing the model to adaptively aggregate neighborhood information across time and produce an interaction-aligned change representation F_c that is robust to registration inconsistencies. Finally, F_c is upsampled and passed through a shallow classifier head to produce the final change map.

B. Semantic Extraction and Relational Encoding

Spatial misalignment can severely disrupt feature learning, hindering the detection of subtle changes. While traditional CNN-based methods are effective at capturing local features, they suffer from limited and rigid receptive fields, making it difficult to model long-range dependencies or adapt to significant pixel displacements and complex deformations. To

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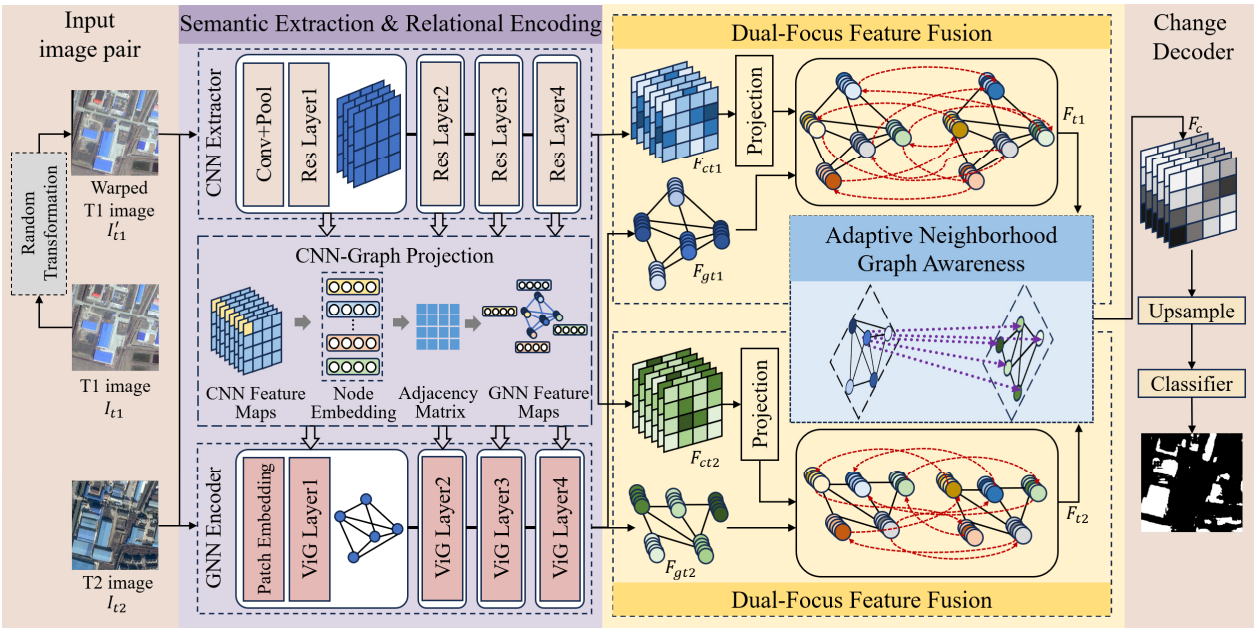


Fig. 2. Overall framework of the ICGINet.

address this challenge, we introduce a dual-branch CNN-GNN backbone: the CNN Extractor captures fine-grained spatial details, while the GNN Encoder models long-range dependencies through flexible message passing, offering a more robust solution to spatial misalignment.

In the CNN extractor, we adopt ResNet-34 [55] as the backbone, which consists of a stem module (a convolution with stride 2 followed by a max-pooling with stride 2), and four residual layers with 3, 4, 6, and 3 basic blocks, respectively. To preserve higher-resolution features, we modify the stride of the first convolution in the first basic block of Layer1 through Layer4 from [1, 2, 2, 2] to [1, 2, 1, 1], while keeping all other components identical to the original design. Given an image pair $I'_{t1}, I_{t2} \in \mathbb{R}^{H\times W\times 3}$, they are processed through a weight-sharing CNN branch, producing feature maps F_{ct1} and F_{ct2} (as shown in Fig. 2), where $F_{ct1}, F_{ct2} \in \mathbb{R}^{C\times h\times w}, h = H/8, w = W/8$.

In the GNN encoder, we employ PyramidViG-S, a pyramid version of Vision GNN (ViG) [39], as the backbone. The architecture consists of a convolutional stem followed by four pyramid stages with 2, 2, 6, and 2 ViG blocks, respectively. Each ViG block integrates a grapher module, which performs dynamic graph convolution to aggregate features from neighboring nodes, and a feed-forward network for feature transformation and diversity enhancement. We adopt the same structure and hyperparameter settings as the original PyramidViG-S implementation.

Given an input image $X \in \mathbb{R}^{H\times W\times 3}$ at a specific timestamp, it is partitioned into $N = (H/4) \times (W/4)$ nonoverlapping patches based on the architecture of the convolutional stem module. Each patch is then encoded into a feature vector $x_i \in \mathbb{R}^D$ by the stem module, forming the set $X = [x_1, x_2, \ldots, x_N]$. To encode positional information, we enhance each vector x_i with a position encoding vector $e_i \in \mathbb{R}^D$. These feature vectors serve as unordered nodes in a graph, denoted as $V = \{v_1, v_2, \ldots, v_N\}$. Following the approach in [39], we construct edges by connecting each node v_i to its $k = 9$ nearest neighbors $\mathcal{N}(v_i)$, forming the edge set $E = \bigcup_{i=1}^{N} e_i$, where e_i represents the edges between v_i and its neighbors $\mathcal{N}(v_i)$. This process yields a graph $G(V, \mathcal{E})$ that encodes spatial dependencies. Our intuition is that semantic entities in the image can be represented as nodes, while their spatial interactions are captured through graph edges. The ViG network then performs relational reasoning to model contextual dependencies and enhance feature representation.

Once the graph is constructed, the ViG backbone processes it through a four-stage hierarchical architecture, where each stage refines node representations via stacked ViG blocks and progressively aggregates contextual information. Given two input images at different timestamps, I'_{t1} and I_{t2}, we extract their corresponding final graph-based feature maps at the end of the ViG backbone, denoted as F_{gt1} and F_{gt2}, respectively. To match the resolution of the CNN branch, these features are subsequently upsampled using two deconvolution layers, resulting in $F_{gt1}, F_{gt2} \in \mathbb{R}^{C\times h\times w}, h = H/8, w = W/8$.

Since the GNN branch constructs graphs from partitioned patches with limited semantic information, we enhance their representation by incorporating intermediate CNN features. Instead of directly linking CNN features to GNN patches, which could introduce semantic inconsistency, we first apply graph projection to transform hierarchical CNN features into a graph-compatible format before integrating them into the GNN branch. Following the approach in [56], we map spatially aligned pixels from F_{cT} into node embeddings, serving as graph nodes. Adjacency matrix is then established by connecting each node to its neighbors based on feature similarity. Next, graph convolution operation update node features via neighborhood aggregation. The refined graph feature maps are then transformed using the adjacency matrix and fused with hierarchical features in the GNN branch for further

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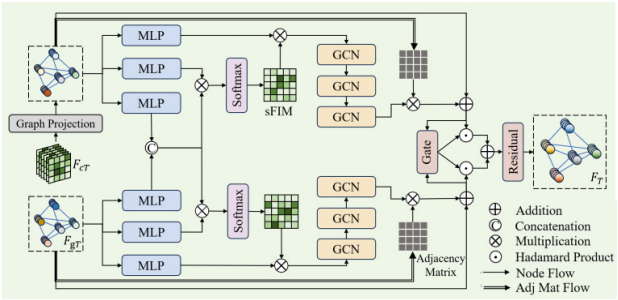


Fig. 3. Overview of the DFFF module.

propagation. The process is formulated as

$$F_{gT}^{i} = \text{concat}\left(F_{gT}^{i}, \left(\text{proj}\left(F_{cT}^{i}\right)\right)\right), \quad i \in \{1,2,3,4\}. \tag{1}$$

The CNN extractor captures high-level semantic features, while the GNN encoder focuses on relational modeling between nodes (entities). Cross-branch fusion enhances node feature semantics in the GNN encoder by integrating complementary information from the CNN extractor.

C. Dual-Focus Feature Fusion

To address the semantic inconsistency between CNN and GNN features, we design the DFFF module to integrate dual-branch features at each temporal phase. Through attention-guided message passing, it achieves effective alignment of heterogeneous representations and enables joint learning of both fine-grained details and structural change features, resulting in more coherent intratemporal features.

The structure of DFFF is illustrated in Fig. 3. In DFFF, F_{cT} at a given timestamp is first processed. To mitigate the semantic gap, F_{cT} undergoes a projection step that refines its representation before graph modeling. Rather than directly linking node features to the adjacency matrix, it is transformed via a graph convolution operation, yielding the graph embeddings $G_c(V_c, A_c)$, where $V_c \in \mathbb{R}^{c\times N}$ encodes node features for N nodes, and $A_c \in \mathbb{R}^{N\times(h\times w)}$ denotes the adjacency matrix. Likewise, F_{gT} is processed through a graph convolution operation to obtain the corresponding graph embeddings $G_g(V_g, A_g)$.

DFFF then facilitates information exchange and feature fusion through attention-guided message passing over both node and edge features. Specifically, separate multilayer perceptrons (MLPs) are used to project V_c into a query graph $V_c^q \in \mathbb{R}^{c/2\times N}$, a key graph $V_c^k \in \mathbb{R}^{c\times N}$, and a value graph $V_c^v \in \mathbb{R}^{c\times N}$. Similarly, V_g is transformed into V_g^q, V_g^k, and V_g^v. Next, we unify V_c^q and V_g^q as

$$V^q = \text{concat}\left(V_c^q, V_g^q\right). \tag{2}$$

Next, a sparse feature interaction matrix (FIM) is constructed to efficiently model the cross-relationships between V_c and V_g. Specifically, for each node in V^q, we compute the similarity scores with all nodes in V_g^k (or V_c^k) and retain only the top-K responses per query. The sparse FIMs are formulated as

$$\text{FIM}_{c\to g}[i,:] = \text{softmax}_K\left(\left(V^q[i]\right)^T V_g^k\right) \tag{3}$$

$$\text{FIM}_{g\to c}[i,:] = \text{softmax}_K\left(\left(V^q[i]\right)^T V_c^k\right) \tag{4}$$

where $\text{softmax}_K(\cdot)$ denotes softmax computed only over the top-K highest scoring positions. As a result, $\text{FIM}_{c\to g}, \text{FIM}_{g\to c} \in \mathbb{R}^{N\times N}$ are sparse matrices with only K nonzero entries per row, significantly reducing computational overhead.

Based on the sparse FIM, dual-focus feature interaction is performed by

$$V_c' = f_{\text{dfff}}\left(V_g, V_c\right) = \left(\text{FIM}_{g\to c} V_c^{vT}\right) \tag{5}$$

$$V_g' = f_{\text{dfff}}\left(V_c, V_g\right) = \left(\text{FIM}_{c\to g} V_g^{vT}\right) \tag{6}$$

where $V_c', V_g' \in \mathbb{R}^{N\times c}$.

The interacted graphs are processed through graph convolution operation to enable cross-branch message passing, enhancing the graph representations. The updated node features are then integrated with the adjacency matrix via batch matrix multiplication and merged with the original features. Then, a gating mechanism is employed to control the contribution weights of the CNN and GNN branches, and the gated features are further integrated with the cross-branch information to obtain the fused representation. The detailed operations are as follows:

$$F_{cT} = f_{\text{GCN}}\left(V_c'\right)^T A_c + F_{cT} = g\left(\text{FIM}_{g\to c} V_c' W_1\right)^T A_c + F_{cT} \tag{7}$$

$$F_{gT} = f_{\text{GCN}}\left(V_g'\right)^T A_g + F_{gT} = g\left(\text{FIM}_{c\to g} V_g' W_2\right)^T A_g + F_{gT} \tag{8}$$

$$G = \sigma\left(W_g \cdot \text{concat}\left(F_{cT}, F_{gT}\right) + b_g\right) \tag{9}$$

$$F_T = \mathcal{R}\left(G \odot F_{cT} + (1-G) \odot F_{gT}\right) \tag{10}$$

where $g(\cdot)$ represents the ReLU activation function, $W_1, W_2 \in \mathbb{R}^{c\times c}$ are the learnable parameters of the graph convolution operation, $G \in \mathbb{R}^{c\times h\times w}$ is a learnable fusion weight map generated via a gating mechanism, $\sigma(\cdot)$ represents the sigmoid activation function, $\odot$ denotes the Hadamard product, and $\mathcal{R}(\cdot)$ denotes a standard residual block composed of convolutional layers and a skip connection that adds the block's input to its output. F_T denotes the enhanced feature representation, which integrates both semantic information and spatial relationships.

D. Adaptive Neighborhood Graph Awareness

Traditional CNNs in image processing primarily rely on fixed-grid neighbor interactions and assume strict spatial correspondence between pixels in bitemporal images. However, this assumption makes CNNs highly sensitive to pixel misalignment, leading to feature inconsistency. Moreover, convolutional kernels are inherently constrained in size and orientation, limiting their ability to capture multidirectional and long-range dependencies. In contrast, graph-based interaction is less reliant on precise pixel-level registration. By representing pixels as graph nodes and modeling their relationships through edges, graph-based methods exhibit greater robustness to spatial misalignment while flexibly capturing multiscale global dependencies. Building on these advantages, we propose the ANGA module, which adaptively adjusts neighborhood perception by incorporating spatial displacement information, enhancing robustness in CD for misaligned RSI.

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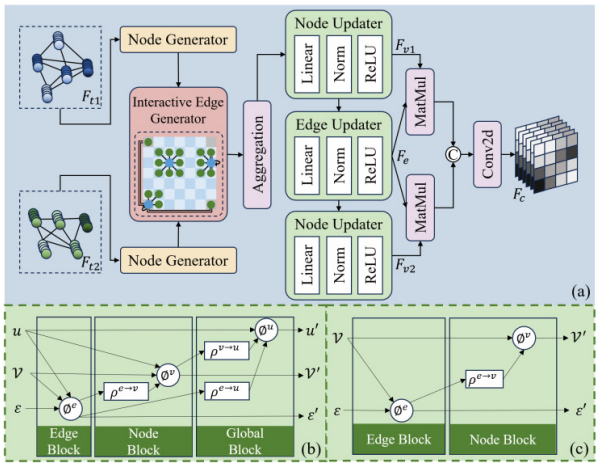


Fig. 4. Overview of the ANGA module. (a) Construction details of the ANGA module, where the blue and green dots in the Interactive Edge Generator denote the nodes of the T1 and T2 time phases, respectively. (b) Illustration of a full GNN block. (c) Illustration of simplified GNN block.

Unlike the intratemporal graph constructed in Section II-B, ANGA constructs a cross-temporal graph. As illustrated in Fig. 4(a), the proposed ANGA module is composed of two main components: interactive graph construction and graph updating.

1) Interactive Graph Construction: Node representations for both input features are first generated using a node generator. Based on these node features, an interactive edge generator constructs cross-temporal edge features, encoding spatial displacement information.

2) Graph Updating: The interactive edge features undergo directional aggregation, followed by node feature refinement of F_{t1} (the displaced temporal feature) via a node updater. Subsequently, an edge updater refines the interactive edge features based on the updated node representations.

Another node updater is then applied again to update the node features of F_{t2}. Finally, the refined edge features are element-wise multiplied with the node features from both temporal phases, followed by concatenation and aggregation to derive interaction-aligned change features. These serve as the input for the classifier to generate the final CD results. The details of the graph construction and updating processes are as follows.

1) *Interactive Graph Construction:* A graph is typically represented as a triplet $G(u, \mathcal{V}, \mathcal{E})$, where u denotes global attributes, $\mathcal{V} = \{\mathrm{v}_k\}_{k=1:N^v}$ is the set of nodes, and $\mathcal{E} = \{(\mathrm{e}_k, r_k, s_k)\}_{k=1:N^e}$ represents the set of edges, with r_k and s_k indicating the indices of the receiving and sending nodes, respectively [57]. Given an input tensor $F_T \in \mathbb{R}^{h\times w\times c}$, we first transform it into a graph $G(u, \mathcal{V}, \mathcal{E})$, where the node set is defined as $\mathcal{V} = \{v_k\}_{k=1}^{hwc}$, and the edge set as $\mathcal{E} = \{(e_k, r_k, s_k)\}_{k=1}^{8hwc}$, with each node connected to its eight nearest neighbors. The node and edge features are then modeled accordingly. In ANGA, rather than constructing separate edge sets for each temporal phase, we employ an Interactive Edge Generator to establish cross-temporal connections that encode spatial displacement information. The process of generating these interactive edge features is illustrated in Fig. 4(a). To ensure structural consistency, special cases at boundaries and corners are explicitly addressed. For boundary nodes, a node at the edge of the graph is treated as one of its own eight neighbors, with its upper and lower neighbors replacing the upper-right and lower-right ones, respectively. Similarly, for corner nodes, the node itself, along with two additional corner nodes, is included in its neighborhood. Finally, a lightweight GNN block is applied to refine the node and interactive edge features, enhancing feature interaction and representation learning.

2) *Graph Updating:* As illustrated in Fig. 4(b), a typical GNN block can be decomposed into three main functional units—an edge block, a node block, and a global block—along with three update functions $(\emptyset^e, \emptyset^v, \emptyset^u)$ and three aggregation functions $(\rho^{e\to v}, \rho^{e\to u}, \rho^{v\to u})$. The overall formulation is as follows [57]:

$$\begin{aligned} \mathrm{e}'_k &= \phi^e\left(\mathrm{e}_k, \mathrm{v}_{r_k}, \mathrm{v}_{s_k}, \mathrm{u}\right) \\ \mathrm{v}'_i &= \phi^v\left(\bar{\mathrm{e}}'_i, \mathrm{v}_i, \mathrm{u}\right) \\ u' &= \phi^u\left(\bar{\mathrm{e}}', \bar{\mathrm{v}}', \mathrm{u}\right) \end{aligned} \tag{11}$$

$$\begin{aligned} \bar{\mathrm{e}}'_i &= \rho^{e\to v}\left(\mathcal{E}'_i\right) \\ \bar{\mathrm{e}}' &= \rho^{e\to u}\left(\mathcal{E}'\right) \\ \bar{v}' &= \rho^{v\to u}\left(\mathcal{V}'\right) \end{aligned} \tag{12}$$

where $\mathcal{E}'_i = \{(\mathrm{e}'_k, r_k, s_k)\}_{r_k=i, k=1:N^e}$, $\mathcal{V}' = \{\mathrm{v}'_i\}_{i=1:N^v}$, and $\mathcal{E}' = \cup'_i\mathcal{E}'_i = \{(\mathrm{e}'_k, r_k, s_k)\}_{k=1:N^e}$. In this model, graph connectivity is determined solely by local neighborhood information, allowing it to be formulated as a Markov random field [58]. The joint distribution of nodes and edges is expressed as

$$P(\mathcal{X}) = \frac{1}{Z}\prod_{\mathrm{v}_i\in\mathcal{Y}}\varphi\left(x_{\mathrm{v}_i}\right)\prod_{(\mathrm{e},r,s)\in\mathcal{E}}\psi\left(x_{\mathrm{v}_r}, x_{\mathrm{v}_s}\right) \tag{13}$$

where $\varphi(x_{\mathrm{v}_i})$ represents the node potential, $\psi(x_{\mathrm{v}_r}, x_{\mathrm{v}_s})$ denotes the edge compatibility between the sender v_s and the receiver v_r, and Z is a normalizer. To approximate the posterior distribution of nodes, inference is performed using the belief propagation method. The message-passing formulation is as follows:

$$m^{(t)}_{r\to s}\left(x_{\mathrm{v}_r}\right) = \varphi\left(x_{\mathrm{v}_s}\right)\psi\left(x_{\mathrm{v}_r}, x_{\mathrm{v}_s}\right)\prod_{k\in\mathcal{N}(\mathrm{v}_s)\setminus\mathrm{v}_r} m^{(t-1)}_{s\to k}\left(x_{\mathrm{v}_s}\right) \tag{14}$$

where $\mathcal{N}(\mathrm{v}_s)$ denotes the neighborhood of v_s, and $m_{r\to s}$ represents the message passed from v_r to v_s. The posterior distribution of the node is given by

$$P^{(t)}\left(x_{\mathrm{v}_i}\right) \propto \varphi_i\left(x_{\mathrm{v}_i}\right)\prod_{k\in\mathcal{N}(\mathrm{v}_i)} m^{(t)}_{ik}\left(x_{\mathrm{v}_i}\right). \tag{15}$$

From (14) and (15), it is evident that the posterior distribution of a node depends only on its node potential function and the edge features of its neighbors, making the global feature u [Fig. 4(b)] redundant. The simplified GNN module [Fig. 4(c)] therefore consists of only the following two components:

$$\mathrm{e}'_k = \phi^e\left(\mathrm{e}_k, \mathrm{v}_{r_k}, \mathrm{v}_{s_k}\right) \tag{16}$$

$$\mathrm{v}'_i = \phi^v\left(\bar{\mathrm{e}}'_i, \mathrm{v}_i\right) \tag{17}$$

where ϕ is computed using an MLP.

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In the graph updating, interactive edge features from different directions are first aggregated through a linear transformation, followed by sequential updates of node and edge representations. This process yields two refined node features, $F_{v1}, F_{v2} \in \mathbb{R}^{c/2\times(h\times w)}$, and an updated set of interactive edge features, $F_e \in \mathbb{R}^{c/2\times(h\times w)}$. These features are then fused through element-wise multiplication and concatenation, generating the final interaction-aligned change representation $F_c \in \mathbb{R}^{c\times h\times w}$, where the fusion operation enables the edge features to guide the node features toward more temporally and spatially consistent alignment under potential misregistration.

E. Prediction Head and Loss Function

After passing through ANGA, the interaction-aligned change feature F_c is upsampled to match the original input resolution. It is then processed by a classifier head consisting of two convolutional blocks, which perform channel transformation followed by sigmoid activation, producing the CD Map $\in \mathbb{R}^{2\times H\times W}$. To optimize the model, we employ the cross-entropy (CE) loss, which quantifies the difference between two probability distributions—in this case, the predicted and ground truth distributions in CD. The CE loss is defined as

$$\text{Loss} = \frac{1}{N}\sum_{i=1}^{N} -\left[y_i \log(\hat{y}_i) + \left(1 - y_i \cdot \log(1-\hat{y}_i)\right)\right] \tag{18}$$

where y_i is the ground truth, $\hat{y}_i$ is the predicted value, and N is the total number of samples.

III. Experiments and Analysis

This section first outlines the experimental setup, including evaluation metrics and implementation details. We then describe the datasets and preprocessing procedures. A comparative analysis with multiple SOTA methods is conducted to demonstrate the superiority of our approach in terms of accuracy, visual quality, and complexity. We also present an ablation study to validate the effectiveness of each component. Furthermore, experiments on real-world scenarios verify the robustness and generalization of our method. Finally, we analyze the impact of perspective transformation on performance.

A. Experimental Setup

1) Evaluation Metrics: To evaluate the effectiveness of our method, we employ four evaluation metrics: Precision (Pre), Recall (Re), F1-score (F1), and Intersection over Union (IoU). These metrics are defined as follows:

$$\text{Pre} = \frac{\text{TP}}{(\text{TP}+\text{FP})} \tag{19}$$

$$\text{Re} = \frac{\text{TP}}{(\text{TP}+\text{FN})} \tag{20}$$

$$F1 = \frac{2\text{Pre}\times\text{Re}}{\text{Pre}+\text{Re}} \tag{21}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP}+\text{FN}+\text{FP}} \tag{22}$$

where true positive (TP) represents correctly detected changes, true negative (TN) denotes correctly identified unchanged areas, false positive (FP) refers to misclassified unchanged areas as changes, and false negative (FN) indicates missed changes. Pre reflects the model's ability to control FP, where a higher Pre indicates fewer false detections. Re measures the model's capability to reduce FN, with a higher Re meaning fewer missed detections. F1, as the weighted harmonic mean of Pre and Re, balances FPs and FNs, where a higher F1 signifies better overall performance. IoU evaluates the overlap between predicted results and ground truth annotations, with a higher IoU indicating better model accuracy.

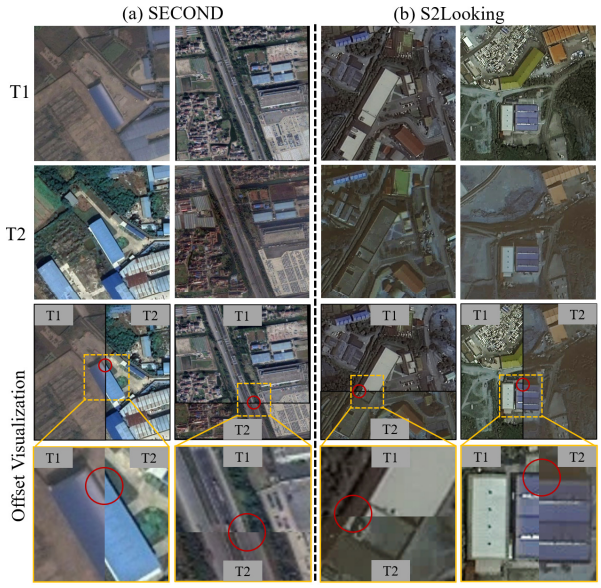


Fig. 5. Representative image samples and their corresponding offset visualization from the two datasets used in the experiments. (a) Exhibits pixel misalignment primarily induced by random perspective transformations and (b) spatial displacement mainly caused by oblique imaging.

2) Implementation Details: All parameters in the network were initialized using a zero-mean Gaussian distribution with a standard deviation of 0.02. Additionally, the network optimization was performed using the SGD optimizer, with a momentum value set at 0.99 and a weight decay of 0.0005.

The training of the network was conducted over 200 epochs with a batch size of 8. A fixed base learning rate of 0.01 was used for the initial stage, followed by a linear decay schedule that gradually reduced the learning rate to 0 over the remaining epochs.

The proposed method, along with all comparative models, was implemented in PyTorch and executed on a single NVIDIA Tesla P100 GPU (16 GB).

B. Datasets

To evaluate the effectiveness of the proposed method, two publicly available and challenging remote sensing CD datasets are used: SECOND [59] and S2Looking [60]. These datasets were selected due to their abundant homologous structures in bitemporal images, which provide a suitable testbed for evaluating our method's robustness to registration error. The diverse land cover types and complex spatial patterns present in these datasets make them particularly appropriate for studying the impact of misalignment in CD tasks (Fig. 5). Table I

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TABLE I
COMPARISON OF INFORMATION FROM THE TWO USED CD DATASETS

Datasets	Image-size	Space-resolution (m/pixel)	Image-pairs	Train/Val/Test	Image-type	Changing-objects
SECOND	512×512	0.5 ~ 3	2968	2077/593/298	aerial images	six classes
S2Looking	1024×1024	0.5 ~ 0.8	5000	3500/500/1000	side-looking satellite images	building

summarizes key characteristics of these two CD datasets. A detailed description of each dataset is provided below.

SECOND dataset is a publicly available semantic CD dataset with images sourced from cities such as Hangzhou, Chengdu, and Shanghai. Each image has a resolution of 512 × 512 pixels and is annotated at the pixel level. SECOND focuses on six primary land cover categories—nonvegetated ground surface, trees, low vegetation, water, buildings, and playgrounds—commonly associated with both natural and human-induced geographic changes. This dataset was selected due to its diverse land cover types and high-resolution imagery, which pose significant challenges for CD in the presence of registration error. Since our task focuses on binary CD, we simplify the dataset by converting all six land cover change categories into two classes: "change" and "no change." This reduces model complexity and enhances generalization capability. Additionally, as SECOND consists of pre-registered image pairs, we follow the approach in [61] and apply random perspective transformations to the pre-change images, introducing slight spatial misalignment to simulate real-world registration error (Fig. 5). A transformation strength parameter θ is introduced to control the degree of distortion during the perspective transformation. In our experiments, we set $\theta = 0.03$ to simulate realistic and moderate misalignment. We provide further analysis of this setting in Section III-F.

S2Looking is a building CD dataset comprising large-scale off-nadir satellite imagery captured at varying off-nadir angles. It consists of 5000 bitemporal image pairs from rural areas worldwide, with over 65 920 annotated change instances. The dataset presents additional challenges for building recognition due to: 1) a wider range of viewing angles; 2) significant illumination variations; and 3) the increased complexity of rural scene characteristics. Although image registration has been applied, the off-nadir imaging geometry and complex mountainous terrain contribute to reduced geolocation accuracy, resulting in residual misalignment of up to eight pixels. To accommodate GPU memory constraints and ensure fair comparisons with other CD methods, we crop all original images into nonoverlapping 512 × 512 patches, yielding 14 000/2000/4000 bitemporal samples for training, validation, and testing, respectively.

C. Comparative Experiment

1) Comparison Algorithms: In this section, we compare ICGINet against 11 SOTA methods, including five CNN- and attention-based methods (FC-Siam-Diff, SNUNet, DMINet, DTCDSCN, and GeSANet), two transformer-based methods (BIT and ICIFNet), two graph-based methods (BGSINet and CFGCN) and two misalignment-handling methods (CASP and ChangeRD).

1) FC-Siam-D (2018) [17]: A Siamese VGG-based model that detects changes by computing absolute feature differences in the encoder.

2) SNUNet (2022) [62]: A densely connected Siamese network based on U-Net++, enhanced with an ensemble channel attention module for multiscale feature extraction.

3) DMINet (2022) [63]: A model that unifies self-attention and cross-attention to optimize global feature distribution and enhance information fusion.

4) DTCDSCN (2020) [64]: A hybrid network combining CD and semantic segmentation to learn discriminative object-level features with dual attention enhancement.

5) GeSANet (2023) [8]: A model leveraging geospatial position matching and cross-dimension exchange mechanisms for refined feature alignment and extraction.

6) BIT (2022) [27]: A transformer-based model that abstracts change regions into high-level semantic tokens for global semantic modeling.

7) ICIFNet (2022) [29]: A hybrid CNN-Transformer model that captures local and global features while enabling spatial-resolution feature interaction via attention.

8) BGSINet (2024) [65]: A graph-based approach that enhances bitemporal feature coupling through semantic interaction in a graph structure space.

9) CFGCN (2024) [45]: A GCN integrating coordinate space and feature interaction, strengthened by boundary-aware and knowledge recall modules.

10) CASP (2025) [37]: A model addressing spatial misalignment by incorporating bidirectional feature alignment and structure-aware fusion for accurate building CD.

11) ChangeRD (2025) [34]: A multitask framework that jointly estimates geometric transformations and change regions to handle misaligned image pairs.

2) SECOND Experiment Results and Analysis: The quantitative results on the SECOND dataset are summarized in Table II. Among the four evaluation metrics, ICGINet achieved the best performance in F1 (67.46%), Re (62.07%), and IoU (50.89%), outperforming the second-best results by 1.26%, 0.67%, and 1.41%, respectively. In terms of Pre, ICGINet ranked second with 73.38%, slightly lower than FE-Siam-D (77.17%). However, it is noteworthy that FE-Siam-D suffers from a significantly lower Re (36.66%), suggesting that its high Pre is achieved at the cost of missing a substantial number of positive samples. Furthermore, we observe that methods explicitly handling misalignment generally achieve higher CD accuracy than those that ignore registration error. Overall, these results demonstrate that ICGINet not only achieves the

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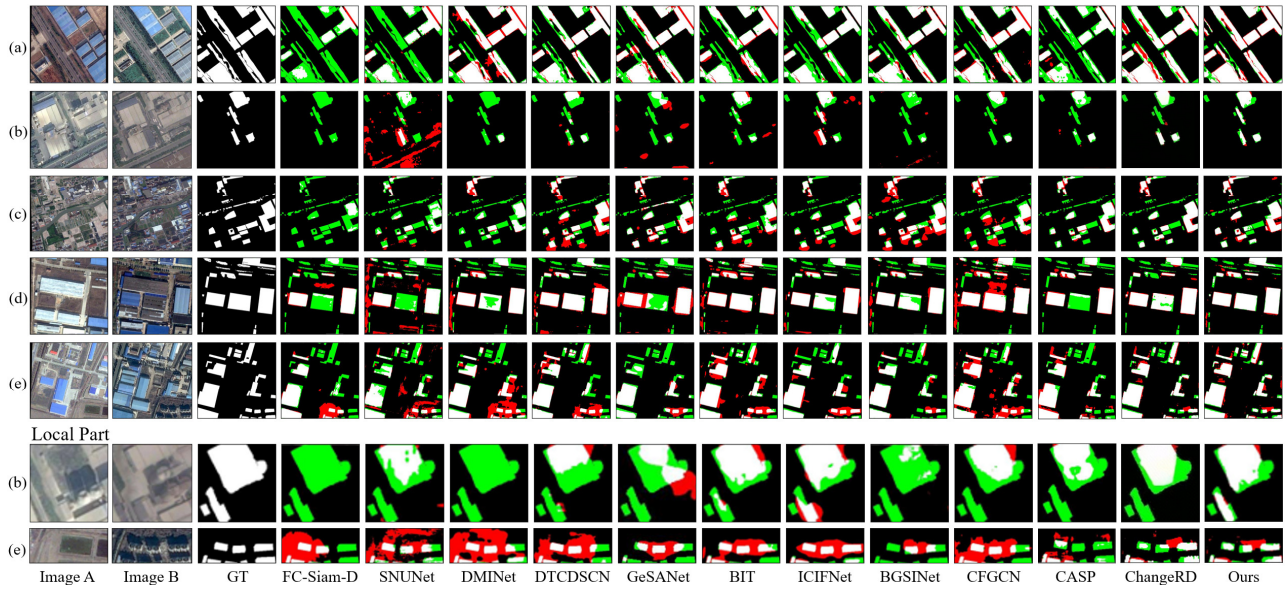


Fig. 6. Visualizations of different CD methods on the SECOND dataset. (a)–(e) Five samples from the test set. The Local Part zooms in on specific regions from two examples to better illustrate the details of the detection results.

TABLE II

AVERAGE QUANTITATIVE RESULTS OF DIFFERENT CD METHODS ON THE SECOND DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN RED, WHILE THE SECOND-BEST RESULTS ARE MARKED IN BLUE

Method	F1 (%)	Pre (%)	Re (%)	IoU (%)
FE-Siam-D	49.71	**77.17**	36.66	33.08
SNUNet	52.41	59.44	46.88	35.51
DMINet	60.90	61.94	59.89	43.78
DTCDSCN	64.83	68.66	**61.40**	48.57
GeSANet	57.43	72.53	47.54	40.28
BIT	64.99	71.42	59.62	48.14
ICIFNet	65.89	73.24	59.88	49.32
BGSINet	61.67	66.84	57.24	44.58
CFGCN	59.96	70.64	52.09	42.82
CASP	64.19	72.85	57.37	47.27
ChangeRD	**66.20**	72.10	61.19	**49.48**
ICGINet	**67.46**	**73.38**	**62.07**	**50.89**

best overall performance on the SECOND dataset but also maintains a better balance between Pre and Re compared to existing methods.

The qualitative visualization results on the SECOND dataset are shown in Fig. 6. To better illustrate detection performance, TPs are marked in white, TNs in black, FNs in green, and FPs in red. As seen in Fig. 6(a)–(e), ICGINet consistently achieves the best overall performance in complex scenarios with diverse land cover types and change patterns. Fig. 6(a) mainly presents a change between bare land and vegetation. Compared to buildings, vegetation exhibits finer and more irregular textures, making it more challenging to detect. ICGINet effectively reduces small-scale FNs and FPs, improves boundary detection accuracy, and minimizes object merging. Fig. 6(b) illustrates a change between vegetation and buildings. ICGINet maintains building integrity, reduces FNs, and avoids FPs caused by shadow interference [see the local region in (b)]. Fig. 6(c) presents an extremely fragmented and complex landscape, where ICGINet demonstrates superior overall performance. Fig. 6(d) and (e) mainly shows changes between bare land and buildings. ICGINet achieves sharper and more precise boundary detection for large buildings while also reducing FNs for small structures. Additionally, in Fig. 6(e), ICGINet effectively distinguishes building shadows from actual structures, ensuring more complete detection of high-rise buildings with significant shadows [see the local region in (e)]. The visualization also reveals that perspective transformations introduce spatial misalignment between bitemporal images, leading all methods to be more prone to FNs and FPs along object boundaries. Among these, FE-Siam-D, SNUNet, DMINet, and GeSANet are most affected, likely due to improper early-stage feature fusion that fails to handle misalignment effectively. In contrast, misalignment-handling methods, particularly ICGINet, are generally less affected by such negative impacts, demonstrating superior robustness to registration errors by significantly reducing both FNs and FPs under spatial shifts.

3) S2Looking Experiment Results and Analysis: The quantitative results on the S2Looking dataset are presented in Table III, demonstrating a significant performance improvement over competing methods. Specifically, ICGINet achieves an F1 of 63.66% and an IoU of 46.69%, outperforming the second-best results by 0.74% and 0.79%, respectively. Although ICGINet ranks second in terms of Pre (70.81%) following FE-Siam-D (75.34%), it should be noted that FE-Siam-D suffers from an extremely low Re (35.12%), resulting in artificially inflated Pre due to a large number of FNs. Likewise, while the Re of ICGINet (57.82%) is slightly lower than that of SNUNet (61.69%), SNUNet exhibits significantly poorer Pre (44.13%), indicating an imbalanced trade-off between the two metrics. Consistent with our obser-

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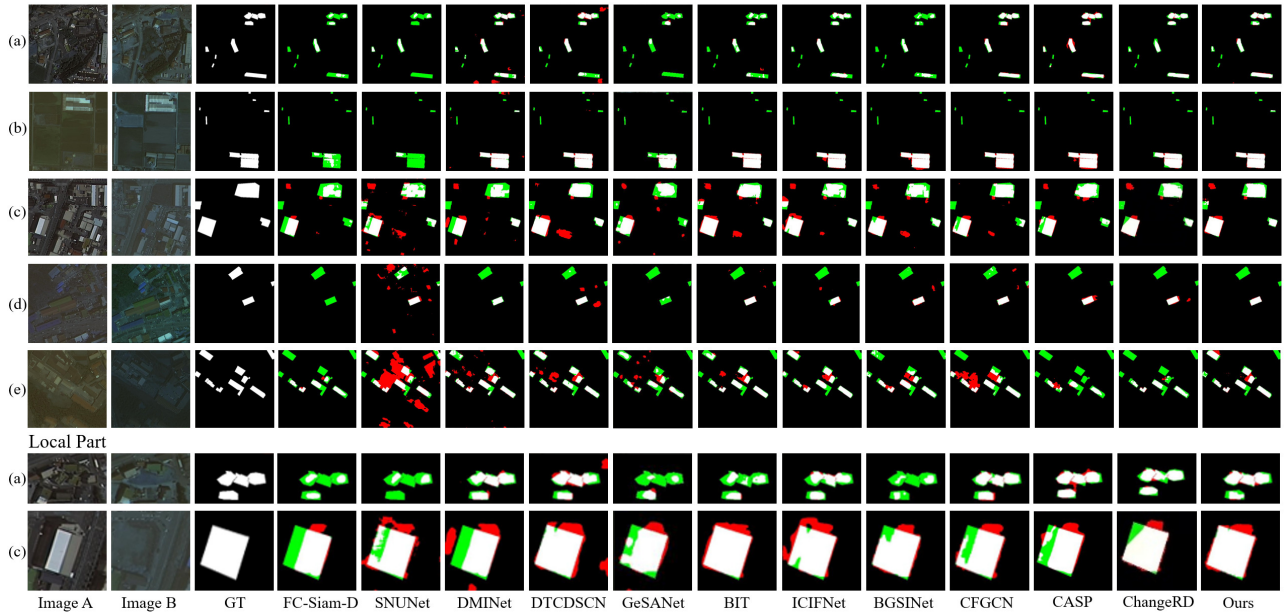


Fig. 7. Visualizations of different CD methods on the S2Looking dataset. (a)–(e) Five samples from the test set. The local part zooms in on specific regions from two examples to better illustrate the details of the detection results.

TABLE III

AVERAGE QUANTITATIVE RESULTS OF DIFFERENT CD METHODS ON THE S2LOOKING DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN RED, WHILE THE SECOND-BEST RESULTS ARE MARKED IN BLUE

Method	F1 (%)	Pre (%)	Re (%)	IoU (%)
FE-Siam-D	47.91	**75.34**	35.12	31.50
SNUNet	51.45	44.13	**61.69**	34.64
DMINet	54.32	63.33	47.56	33.45
DTCDSCN	62.60	68.29	57.77	45.61
GeSANet	52.29	61.93	45.24	30.85
BIT	61.98	70.79	55.12	44.91
ICIFNet	58.77	67.19	52.22	41.61
BGSINet	55.87	65.94	48.46	38.76
CFGCN	60.47	64.35	57.03	43.33
CASP	60.96	65.29	57.17	43.85
ChangeRD	**62.92**	70.69	56.71	**45.90**
ICGINet	**63.66**	**70.81**	**57.82**	**46.69**

vations on the SECOND dataset, methods explicitly addressing misalignment also achieve higher CD accuracy on S2Looking, further reinforcing the importance of handling geometric inconsistencies. Overall, ICGINet achieves the best overall performance on the S2Looking dataset by maintaining a superior balance between Pre and Re, thereby demonstrating its robustness and generalization capability.

The qualitative visualization results on the S2Looking dataset are shown in Fig. 7. As seen in Fig. 7(a)–(e), the oblique imaging angle exposes building facades and shadows, posing a significant challenge for CD. However, ICGINet effectively mitigates FNs and FPs, achieving the best overall performance. In Fig. 7(a), the buildings in the upper part of the image exhibit visible side facades and shadows, often causing detection errors in competing methods, whereas ICGINet accurately identifies the changed structures [see local region in (a)]. Similarly, in Fig. 7(c), the buildings in the lower-left part of the image demonstrate this issue—methods like GeSANet and CFGCN fail to detect parts of the tilted structure, while ICGINet provides a more complete and precise detection [see local region in (c)]. For smaller and fragmented buildings, ICGINet also reduces FNs [Fig. 7(b), (d), and (e)], ensuring structural integrity while minimizing object merging and over-segmentation. Overall, ICGINet demonstrates strong registration robustness, significantly reducing errors even in the presence of spatial misalignment caused by side-view imaging.

4) Efficiency Comparison: We evaluate the efficiency of different methods using the number of trainable parameters (Params) and floating-point operations (FLOPs), which respectively reflect model size and computational cost. Given an input image pair $I_1, I_2 \in \mathbb{R}^{3\times512\times512}$, the results are summarized in Table IV. Compared with most existing methods, ICGINet maintains a relatively low model complexity, with only 9.6M parameters and 96.3G FLOPs, while delivering top performance in terms of F1 and IoU on both datasets. While some lightweight models achieve lower Params or FLOPs, their accuracy tends to be modest, whereas other methods with higher accuracy often involve increased computational demands. In contrast, ICGINet strikes a more favorable balance.

D. Ablation Experiments

ICGINet employs a dual-branch CNN-GNN feature extraction and rational encoding network to capture enhanced features that combine high-level semantic abstraction with global dependencies. The DFFF module then facilitates feature interaction and information propagation through attention-guided graph communication. Finally, the ANGA module

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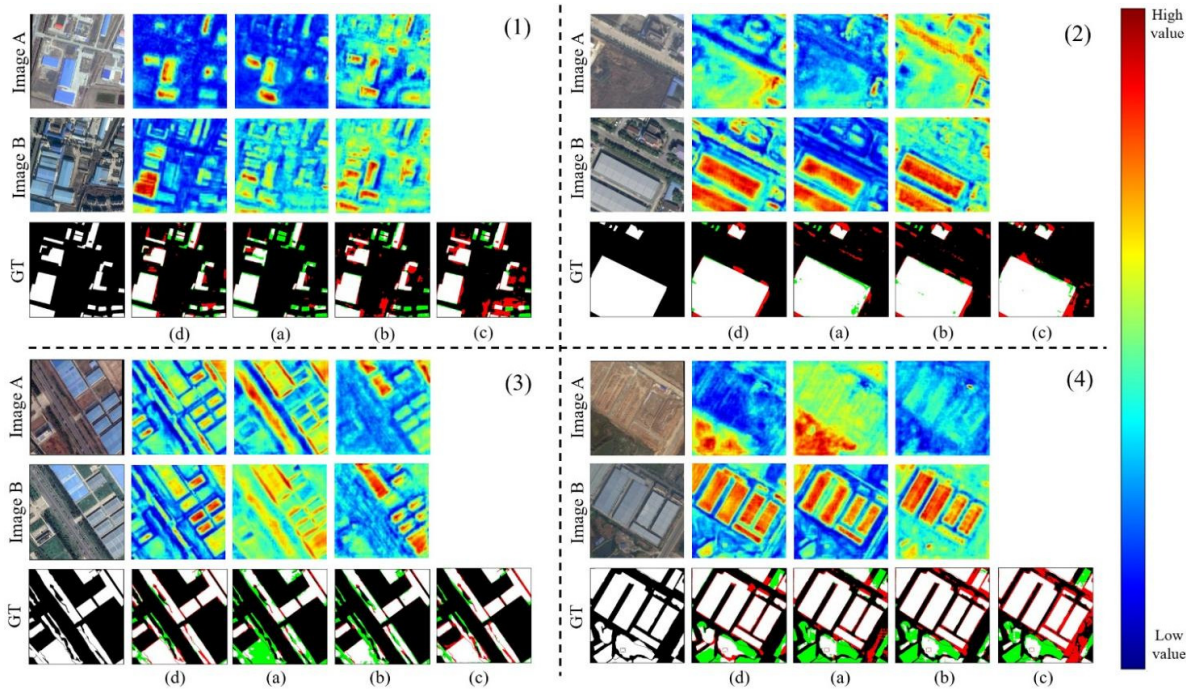


Fig. 8. (a)–(d) Visualizations of heatmaps and CD predictions for different ablation configurations on four samples from the SECOND dataset.

TABLE IV
EFFICIENCY AND PERFORMANCE COMPARISON OF EACH METHOD

	Params (M)	FLOPs (G)	SECOND		S2Looking	
			F1 (%)	IoU (%)	F1 (%)	IoU (%)
FE-Siam-D	1.350	18.907	49.71	33.08	47.91	31.50
SNUNet	12.035	219.333	52.41	35.51	51.45	34.64
DMINet	6.242	59.494	60.90	43.78	54.32	33.45
DTCDSCN	31.257	52.895	64.83	48.57	62.60	45.61
GeSANet	36.503	8.621	57.43	40.28	52.29	30.85
BIT	3.496	42.525	64.99	48.14	61.98	44.91
ICIFNet	23.843	25.410	65.89	49.32	58.77	41.61
BGSINet	8.831	82.543	61.67	44.58	55.87	38.76
CFGCN	13.583	176.328	59.96	42.82	60.47	43.33
CASP	12.737	10.515	64.19	47.27	60.96	43.85
ChangeRD	29.468	73.096	**66.20**	**49.48**	**62.92**	**45.90**
ICGINet	9.602	96.252	**67.46**	**50.89**	**63.66**	**46.69**

TABLE V
EVALUATION OF DIFFERENT MODULES ON THE SECOND DATASET

Methods	dual-branch	DFFF	ANGA	F1 (%)	Pre (%)	Re (%)	IoU (%)
(a)	×	×	√	63.71	71.88	57.20	47.96
(b)	√	×	√	64.65	70.17	59.94	48.65
(c)	√	√	×	65.16	69.91	61.03	48.94
(d)	√	√	√	67.46	73.88	62.07	50.89

refines change features with registration robustness by aggregating interactive edge features and updating node features. In this section, we conduct ablation experiments on the SECOND dataset to evaluate the effectiveness of different components in ICGINet. The schematic design and results are presented in Table V: (a) uses only a conventional CNN branch for feature extraction, without the dual-branch design or feature fusion; (b) employs the dual-branch feature extraction network but replaces the DFFF module with a simple concatenation operation for feature aggregation; (c) incorporates both the dual-branch network and the DFFF module for feature fusion but omits the ANGA module, instead using simple concatenation to obtain change features; and (d) ICGINet. To provide a more intuitive comparison, Fig. 8 visualizes the CD results of four sample cases across configurations (a)–(d), along with heatmaps of bitemporal feature representations generated after inter-temporal fusion and before ANGA, which operates on these fused bitemporal representations to generate change features.

1) Dual-Branch Design Enhances Long-Range Dependencies and Feature Representation: As shown in Table V (a) and (d), ICGINet achieves F1, Pre, Re, and IoU improvements of 3.75%, 2.00%, 4.87%, and 2.93%, respectively, compared to the single-branch feature extraction network. This highlights the role of the GNN branch in strengthening feature representation. A more intuitive comparison can be observed in the feature heatmaps of Fig. 8(a) and (d). Without the GNN branch, the extracted features exhibit weaker and less focused activations, leading to reduced object integrity. Similarly, the CD results of (a) and (d) show that the absence of the GNN branch results in a higher rate of missed detections, negatively impacting object completeness. These visualizations further confirm that the dual-branch network significantly improves feature extraction capabilities. The results demonstrate that, while the CNN branch captures high-level semantic features,

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incorporating a GNN branch effectively models long-range dependencies and supplements global contextual information, leading to more robust feature learning [39].

2) Effective Feature Fusion is Essential for Enhanced Representation: CNN-based features primarily focus on local semantic information, while GNN-based features capture global relationships. Given the differing focus of each branch, an appropriate feature fusion strategy is crucial for effectively integrating complementary information. Compared to the simple concatenation approach in Table V (b), incorporating the DFFF module in (d) leads to F1, Pre, Re, and IoU improvements of 2.81%, 3.71%, 2.13%, and 2.24%, respectively. This demonstrates the effectiveness of DFFF in facilitating cross-branch interaction and joint representation learning. A more intuitive comparison can be observed in the feature heatmaps of Fig. 8(b) and (d). Without DFFF, the naive fusion approach results in poor information interaction, causing the network to focus more on background regions while exhibiting weaker responses to critical areas, leading to blurred details. The CD results of (b) and (d) further confirm that inadequate feature aggregation increases the likelihood of missed and false detections. These visualizations highlight the importance of DFFF in effectively integrating complementary features from different branches, enabling better interaction and representation learning across features with varying semantic focus.

3) ANGA is Crucial for Enhancing Alignment Robustness: As shown in Table V (c) and (d), incorporating the ANGA module leads to F1, Pre, Re, and IoU improvements of 2.30%, 3.97%, 1.04%, and 1.95%, respectively, highlighting its significant impact on overall performance, particularly in boosting Pre. A more intuitive comparison can be observed in the CD results of Fig. 8(c) and (d). Without the ANGA module, the model is more prone to false detections, especially along object boundaries, resulting in blurred boundaries and severe merging or adhesion between objects. The visualizations in Fig. 8 further illustrate that ANGA effectively mitigates boundary misalignment caused by registration error, enhancing the model's robustness to spatial discrepancies in CD.

To rigorously assess whether the enhanced robustness of ANGA in registration is genuinely attributed to its dynamic edge interaction mechanisms rather than merely the increase in learnable parameters, we designed a dedicated experiment to evaluate its resistance to misalignment errors, as summarized in Table VI. Specifically, we compare four configurations that maintain an identical number of learnable parameters: (a) represents the complete version of our ANGA module, which integrates the Interactive Edge Generator and employs a neighborhood size of $k = 8$; (b) is a variant of this design with a reduced neighborhood size ($k = 4$), allowing for a comparative analysis of neighborhood scale effects; (c) modifies the original design 1) by replacing the interactive edge generator with an Independent Edge Generator, ensuring that each timestamp maintains its own distinct edge set rather than leveraging cross-temporal interactive edges; and (d) retains the interactive edges but omits the fusion step, directly concatenating F_{v1} and F_{v2} to obtain change features. These experiments validate that ANGA's effectiveness in handling registration error stems from its ability to dynamically update interactive edges and node information, incorporating spatial neighborhood awareness, rather than simply increasing model complexity. As shown in Table VI, increasing the number of neighboring nodes [comparing (a) and (b)] results in F1, Pre, Re, and IoU improvements of 0.72%, 1.68%, 0.01%, and 1.07%, respectively. This indicates that ANGA enhances the perception of spatial neighborhood relationships, particularly improving Pre by leveraging interaction-based feature learning. Similarly, when comparing (a) and (c), the use of interactive edges leads to notable gains of 1.82% in F1%, 3.12% in Pre, 0.86% in Re, and 1.75% in IoU. This demonstrates that ANGA effectively captures and propagates alignment information through interactive edges, mitigating the adverse impact of registration error on CD accuracy. Comparing (a) and (d), consistent performance drops are observed when the interactive edge features are not fused, further confirming the importance of this component in enhancing alignment-aware representation and mitigating registration-induced pseudo-changes.

TABLE VI
EVALUATION OF ANGA'S ROBUSTNESS TO REGISTRATION ERROR ON THE SECOND DATASET

Methods	design of ANGA	F1 (%)	Pre (%)	Re (%)	IoU (%)
(a)	ANGA-base	67.46	73.88	62.07	50.89
(b)	ANGA-k4	66.75	72.20	62.06	49.82
(c)	ANGA-indep	65.64	70.77	61.21	49.14
(d)	ANGA-nofuse	66.09	72.10	60.99	49.35

E. Comparison of Real-World Change Scenarios

In CD tasks, diverse and realistic data are crucial for evaluating a model's generalization ability and robustness to complex scene variations. To this end, several representative real-world remote sensing samples were selected from different locations in China using Google Earth, spanning two distinct time periods (2018–2024). The dataset encompasses typical land cover changes, including transitions among buildings, nonvegetated ground surfaces, and vegetation. In addition, the collected bitemporal images exhibit inherent misalignments, reflecting practical challenges in real-world CD scenarios. A comparative evaluation against five SOTA models was conducted to assess the proposed method's performance under such conditions. To address the challenges posed by diverse land cover types and complex structural variations in real-world scenarios, we employed the model weights that achieved the best performance on the SECOND dataset for further evaluation. The SECOND dataset encompasses a wide range of real-world change types, offering strong representativeness and serving as a solid benchmark to assess the adaptability of models across various scenes. Without any additional fine-tuning, the pretrained model was directly applied to real RSI to evaluate its robustness and generalization capability under realistic conditions.

The quantitative results on six real-world remote sensing samples are summarized in Table VII. Among the four evaluation metrics, ICGINet achieved the best performance across

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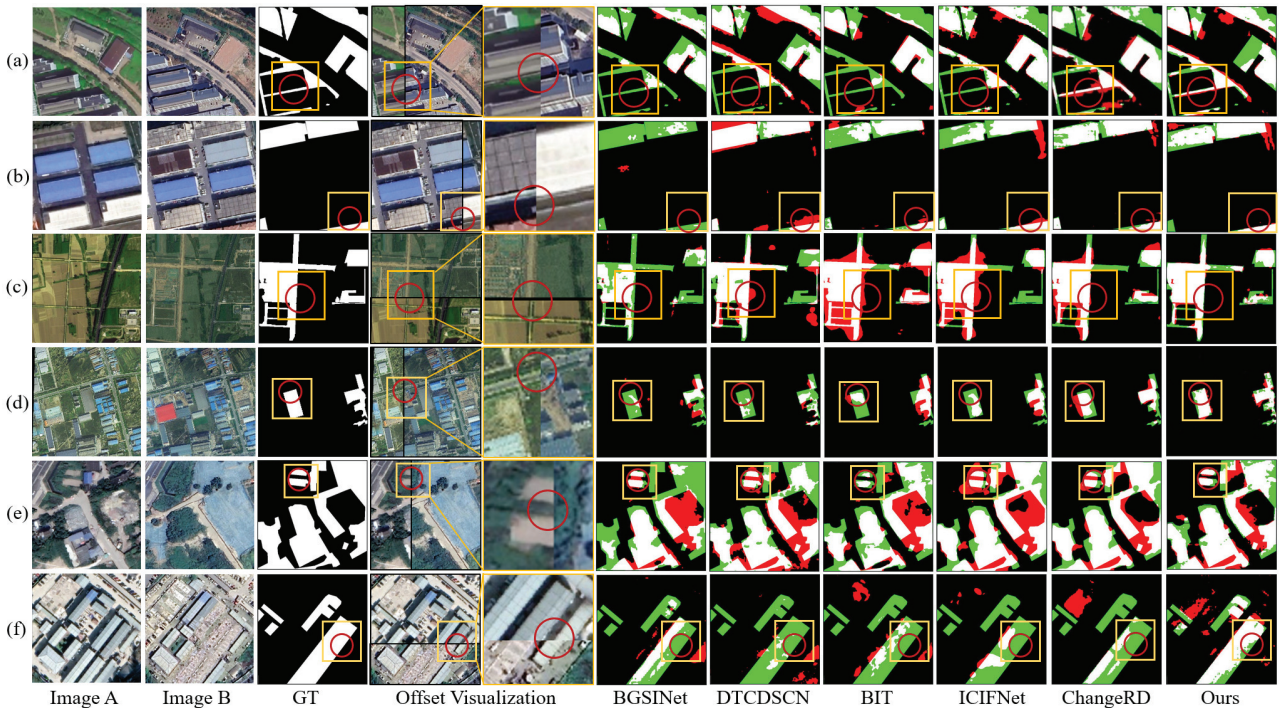


Fig. 9. Qualitative comparison of the prediction results on the six real-world remote sensing samples (a)–(f).

TABLE VII
QUANTITATIVE RESULTS OF DIFFERENT CD METHODS ON THE REAL-WORLD REMOTE SENSING SAMPLES

Method	F1 (%)	Pre (%)	Re (%)	IoU (%)
BGSINet	49.16	86.20	36.34	34.70
BIT	66.75	77.56	62.96	50.51
DTCDSCN	71.19	80.15	67.93	56.53
ICIFNet	71.82	79.52	66.54	56.27
ChangeRD	79.04	84.57	74.54	65.83
ICGINet	81.22	87.27	75.95	68.38

all indicators, surpassing the second-best method by margins of 2.73% in F1%, 0.45% in Pre, 5.08% in Re, and 3.59% in IoU, respectively. The corresponding qualitative results are presented in Fig. 9. To better visualize the impact of pixel misalignment on CD accuracy, areas with noticeable misregistration are highlighted using yellow boxes and red circles. Overall, ICGINet demonstrates generally improved robustness and detection accuracy under imperfect image registration, producing more reliable results compared with baseline methods. In Fig. 9(a), tilted buildings and shadow interference pose challenges for most baseline approaches in detecting subtle changes; while ChangeRD is able to capture these changes, it also introduces minor FPs, whereas ICGINet yields comparatively more precise results. In Fig. 9(b), under the combined influence of pixel displacement and shadows, several methods tend to produce spurious detections along change boundaries. By contrast, ICGINet benefits from robust feature learning, which helps suppress FPs and maintain boundary accuracy. In Fig. 9(c) and (d), where slight road misalignments occur, ICGINet achieves relatively more accurate boundary detection. In the more complex case shown in Fig. 9(e), ICGINet reduces FPs caused by the bluish ground surface while preserving precise boundary detection. Finally, in Fig. 9(f), although minor disturbances such as vehicles and ground debris give rise to some FPs, ICGINet is still able to identify most of the true change regions effectively. It is also worth noting that the performance of all methods appears less satisfactory in these real-world cases, which is understandable given the complexity of practical scenarios. Factors such as long temporal intervals, seasonal and illumination variations, and differences in sensor characteristics often lead to appearance inconsistencies that do not correspond to actual changes, thereby increasing the overall difficulty of detection.

F. Analysis of Perspective Transformation and Image Misalignment

To simulate realistic spatial misalignment in pre-registered bitemporal image pairs, we apply a random perspective transformation to the pre-registered dataset. Specifically, we introduce spatial distortion by perturbing the four corner points of each pre-change image and estimating a homography matrix to perform a global perspective warp.

A transformation strength parameter θ is introduced to control the distortion level. Larger θ values result in stronger geometric deformations and greater spatial displacement between the paired images. Fig. 10 shows qualitative examples under $\theta = 0.01$, 0.03, and 0.05. While increasing θ clearly amplifies the misalignment, random corner selection also introduces sample-wise variation, mimicking the diversity of real-world registration errors.

We further quantify the distortion by computing the center shift and average pixel-wise misalignment on the SECOND test set. As shown in Table VIII, $\theta = 0.03$ provides a moderate level of distortion, aligning well with typical registration errors

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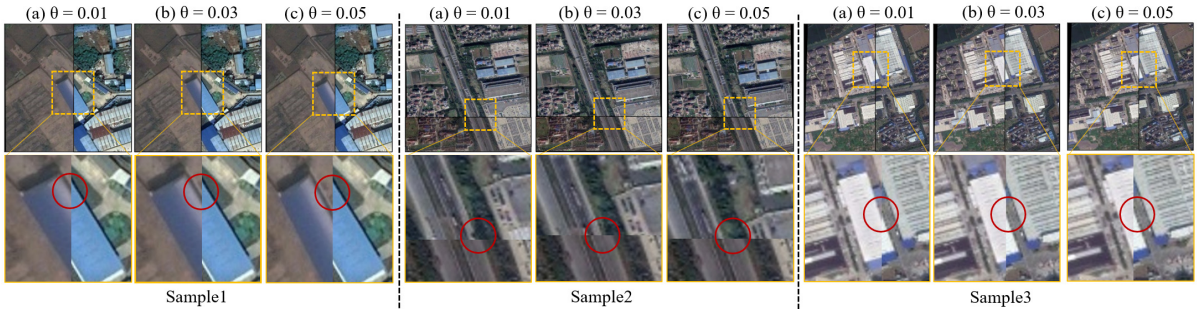


Fig. 10. Visual comparison of perspective transformation with varying θ settings (a)–(c) on random samples.

TABLE VIII
QUANTITATIVE EVALUATION OF SPATIAL MISALIGNMENT INTRODUCED BY PERSPECTIVE TRANSFORMATION WITH DIFFERENT θ VALUES

θ	Center Shift (px)	Avg Misalignment (px)
0.01	4.82	4.76
0.03	12.60	12.59
0.05	20.45	20.46

in remote sensing. Therefore, we adopt $\theta = 0.03$ in our main experiments.

IV. CONCLUSION

In this article, we propose ICGINet, an intratemporal and cross-temporal graph interaction network designed to achieve registration-robust CD in high-resolution RSI. To address the challenges posed by spatial misalignment, ICGINet constructs flexible graph interaction structures across both intratemporal and cross-temporal domains. Specifically, a dual-branch CNN-GNN backbone extracts complementary local semantics and global relational features, while the proposed DFFF module adaptively aligns and integrates intratemporal features through attention-guided graph message passing. Furthermore, the ANGA module dynamically builds and updates cross-temporal graphs to capture spatially displaced neighborhood interactions, significantly mitigating registration-induced inconsistencies. Extensive experiments on two challenging datasets demonstrate that ICGINet consistently outperforms 11 state-of-the-art methods in both quantitative qualitative results, highlighting its strong robustness to spatial misalignment. In the future, we plan to further enhance the model's computational efficiency through lightweight architecture design and more efficient graph construction strategies.

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